

Questions Regarding the DRAGON Code Release

Related to the paper “*DRAGON: Discrete-Fractional Graph Neural Operators*”

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I am very interested in reproducing the results of the DRAGON paper and studying the methodology further. While going through the GitHub repository, I encountered some ambiguities and missing details. Below is a structured list of questions that I kindly ask the authors to clarify.

1. Environment and Dependencies

- Could you please provide the exact `environment.yaml` or `requirements.txt` file (Python, CUDA, PyTorch, PyG, torchdiffeq, torchcde, torchfde, etc.)?
- Which commit/version of `torchfde` was used?

2. Missing Files and Block Types

- Several referenced files are missing: `base_classes.py`, `best_params.py`, `heterophilic.py`, `regularized_ODE_function.py`. Will these be released or have they been renamed?
- Configurations mention `ConstantODEblock_FRAC_MULTI_ORDER`, but only constant fraction-order blocks appear in code. Which block names are correct?

3. Datasets and Preprocessing

- For Twitch, which subsets (e.g., DE, ENGB, etc.) or versions did you use?
- What preprocessing steps (normalization, self-loops, isolated nodes) were applied?
- Are train/validation/test splits fixed or random? If fixed, could you provide split files or seeds?

4. Hyperparameters and Training Protocol

- Could you share the best hyperparameters per dataset (lr, wd, dropout, hidden size, epochs, scheduler, patience, gradient clipping)?
- How many seeds/runs were averaged, and which seeds?
- Was any learning rate scheduler used?

5. Fractional / Multi-Order Settings

- Are fractional orders α fixed or trainable? How are they initialized and constrained?
- How are multi-order weights normalized (softmax, simplex, etc.)?
- Which solver was used (step size, rtol/atol, adjoint method)?
- Any numerical stability tricks (regularization, smoothing, clipping)?

6. Evaluation Protocol

- Which loss function was used (cross-entropy, label smoothing, weights)?
- Which metrics (accuracy, F1, etc.), and on validation or test?
- Any post-processing or calibration applied?
- Could you share reference outputs/logs for verification?